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Search for a doubly charged Higgs decaying to muons or electrons: Prospects for CMS Run III

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On behalf of the IMSA-CMS group

Goals of This Talk

- Explore the doubly charged Higgs (H++)
  - Models
  - Production Mechanisms
- Signal
  - H++ event signature
  - Signal channels
- Background
  - Estimation
- Estimated limits on the H++
- Future Work
- Automatic Datacard Generation
- Punzi Sensitivity Approximation

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Doubly Charged Higgs

- Left-Right Symmetric model
  - Parity violation in the weak sector
- The LRS model predicts a Higgs triplet
  - Neutral Higgs (not the SM Higgs)
  - Singly Charged Higgs
  - Doubly Charged Higgs

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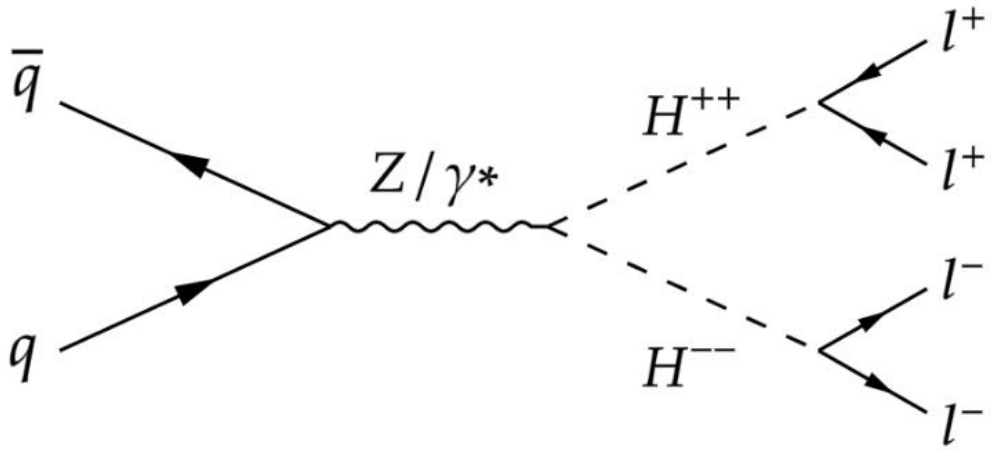
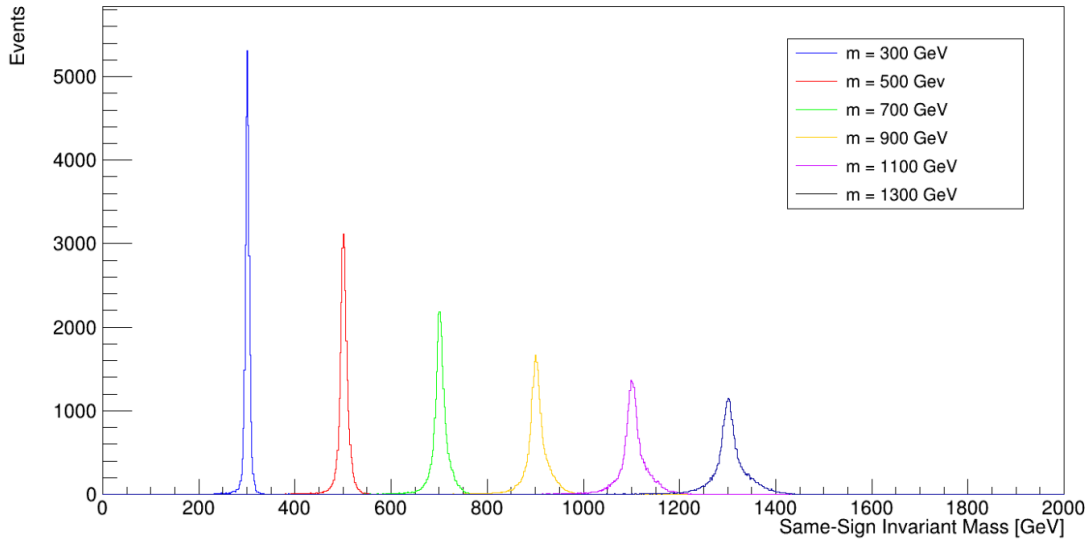
Doubly Charged Higgs Production

- A pair of oppositely charged doubly charged Higgs can be produced by a Z boson or electro-weak production
- Higgs couples with any massive particle



Event Signature: Same Sign Dileptons (SSDL)

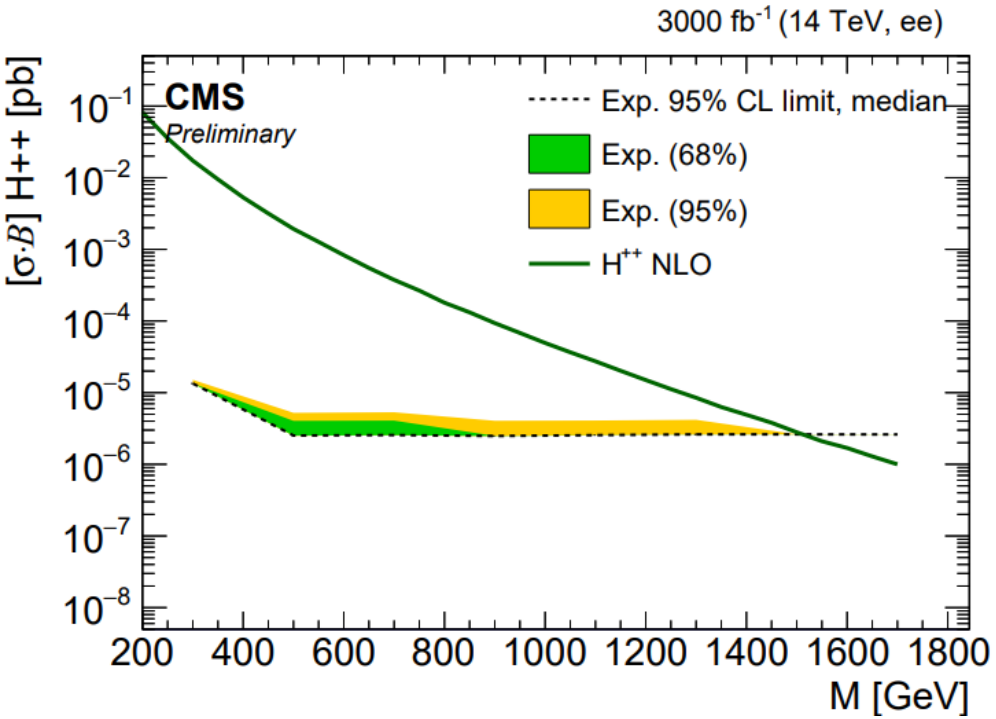
- The H++ can produce a pair of leptons with the same charge
- A H++ event will produce a total of 4 leptons split into same-sign pairs
- This is a very rare signature in pp collisions

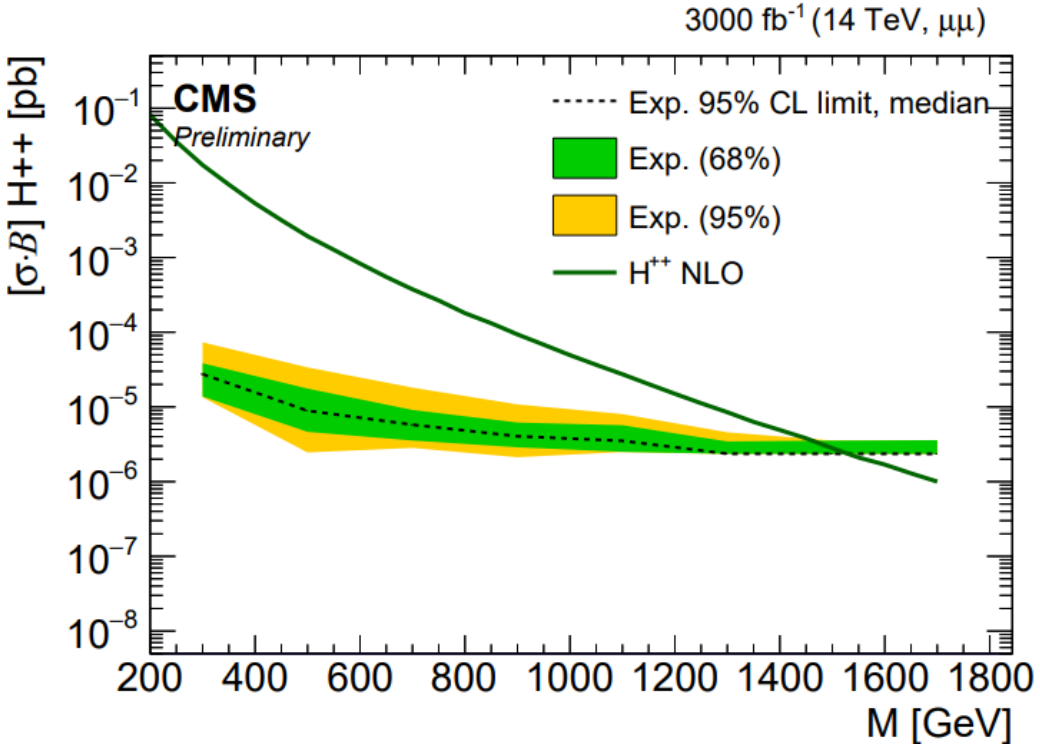


A Different Approach to Estimating Background

- Our background has an incredibly small cross section and comes from primarily from systematic errors
  - Charge mismeasurement and photon fakes combined account for 75 ± 16.324%
- In order to estimate such background a different method is required from the usual
  - The efficiency is too low
  - Monte Carlo (count how many events pass cuts) is worse for low frequency phenomenon
- Numerical estimate

Estimated Limits





- Estimated limits for the HL-LHC
- Scaling by 1/√(luminosity)
- Estimated Run III limits:
  - 1300 GeV, ee
  - 1325 GeV, uu

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Automatic Datacard Generation

- Datacards
  - Encodes observed and expected events after event selection
  - Distinguishes between signal vs. background
  - Manual setting of datacards
    - Human error
    - Time consuming
- Developed framework for automatic datacard generation
- Created H++ analysis object
  - Channel
    - Make a txt file with channel name
    - Process
      - Yields + systematics
- Higgs Combine Tool
  - Runs thousands of pseudoexperiments based off datacards
  - Returns the limit on the Higgs

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Example Datacard

```
imax 4 number of bins
jmax 5 number of processes minus 1
kmax 0 number of nuisance parameters
--------------------------------------------------------------------------------------------------------------------------------
bin          Lep2        Lep3        Lep4        Lep5
observation  43.4204533  43.4204533  43.4204533  43.4204533
--------------------------------------------------------------------------------------------------------------------------------
bin          Lep2        Lep2        Lep2        Lep2        Lep2        Lep2        Lep3        Lep3        Lep3        Lep3        Lep3        Lep3
process      Higgs       DYMuons     WZ3LNu      DYElectrons ZZ          TTBar       Higgs       DYMuons     WZ3LNu      DYElectrons ZZ          TTBar
process      0           1           2           3           4           5           0           1           2           3           4           5
rate         0.0052796301 2942.016539 4.519904887 9.726097434 1.076796211 609.65      0.1658915382 478.7322276 4.842755236 1.015842367 0.7487769504 0
--------------------------------------------------------------------------------------------------------------------------------
```

```
    HiggsPlusPlusAnalysis analysisObj;
    auto channels = analysisObj.getChannels();

  for(auto channel : channels)
    {
        channel->makeDatacard(channel);
    }
    return 0;
{
    auto processName=Utility::substitute(process->getName()," ", "_");
    datacard<< std::setw(20) << std::left << processName;
}

datacard<<"\n";
datacard<< std::setw(20) << std::left << "rate";

for (auto process : channel->getProcesses())
{
    auto yield=std::to_string(process->getYield("Invariant Mass"));
    datacard<< std::setw(20) << std::left <<yield;
}
```

```
void Channel::makeDatacard(std::shared_ptr<Channel> channel)
{
    std::string channelName = channel->getName();
    std::string filename=channelName+".txt";
    std::ofstream datacard (filename);
    if(!datacard)
    {
        throw std::runtime_error("Unable to create file");
    }
    datacard << "\nimax    1 number of bins\n";
    datacard << "jmax    _______ number of processes minus 1\n";
    datacard << "-----------------------------------------------
    datacard <<"bin";

    for (auto i = 0; i < 5; i++)
    {
        datacard <<std::setw(20) << std::left << channelName;
    }

    datacard<< std::setw(20) << std::left << "process"<< std::left;
```

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Definition: Sensitivity

- Helps us determine what point to employ a cut
  - Reduce background to better identify signal
  - Can’t cut out too much signal in the process
- Example for H++
  - Background: QCD, Drell-Yan, Ttbar, BLAHHHHHHHH

  - PUT PICTURE FOR QUALITATIVE ANALYSIS OF CUT SELCECTION

Sensitivity Estimation

- Our group's selection cuts were based on the standard significance = S/sqrt(B) formulation
- Flaws
  - Depends on a-priori expectations of limit settings
- Punzi Criterion
  - Independent of a-priori expectations to assert that a discovery has been made
  - Derived simple approximation of Poisson counts with background

Derivation: Definitions

- X – Observed variable, H_0 – distribution of X
- H_m is Poisson distribution of X with mean equal to the number of B+S_m
  - S_m new signal, m the unknown free parameter of the signal
- Test H_0 by selecting range of X that helps us reject H_0
  - Critical Region
  - Alpha is the significance level of rejecting H_0 when it is indeed true
  - Alpha = 5 sigma

Derivation: Criteria

- Independent of choice of metric
  - In both observable and parameter space
- Should not require a choice of priors and independent of expectation of signal
  - Priors are assumptions we make about signal

$$P(n|H_m) = e^{-B-S_m} \cdot (B + S_m)^{n/n!}$$ and $$P(n|H_0) = e^{-B} \cdot B^{n/n!}$$

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Derivation: Some Assumptions

- S_m is non-zero
  - n>n_min
  - S_m >S_min

Derivation

- Using a gaussian (ask dr dong if should be capitalized, weird convo) of the Poisson, we can state that

$$S_{min} = a\sqrt{B} + b\sqrt{B + S_{min}} \implies S_{min} = \frac{b^2}{2} + a\sqrt{B} + (\frac{b}{2})(b^2 + 4a\sqrt{B} + 4B)$$

- Assume efficiency of signal cuts are independent of m, that is $S_m(t) = e(t) \cdot L \cdot \sigma_m$ and we aim to maximize sensitivity by minimizing sigma_m

$$e(t) \cdot \frac{L}{\frac{b^2}{2} + a\sqrt{B}} + (\frac{b}{2})(b^2 + (b^2 + 4a\sqrt{B} + 4B(t))$$

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Derivation and Comparison

- Setting a = b = 5sigma, we arrive at

$$\frac{S}{2.5 + \sqrt{B}}$$

- Consider the common sensitivity measures:

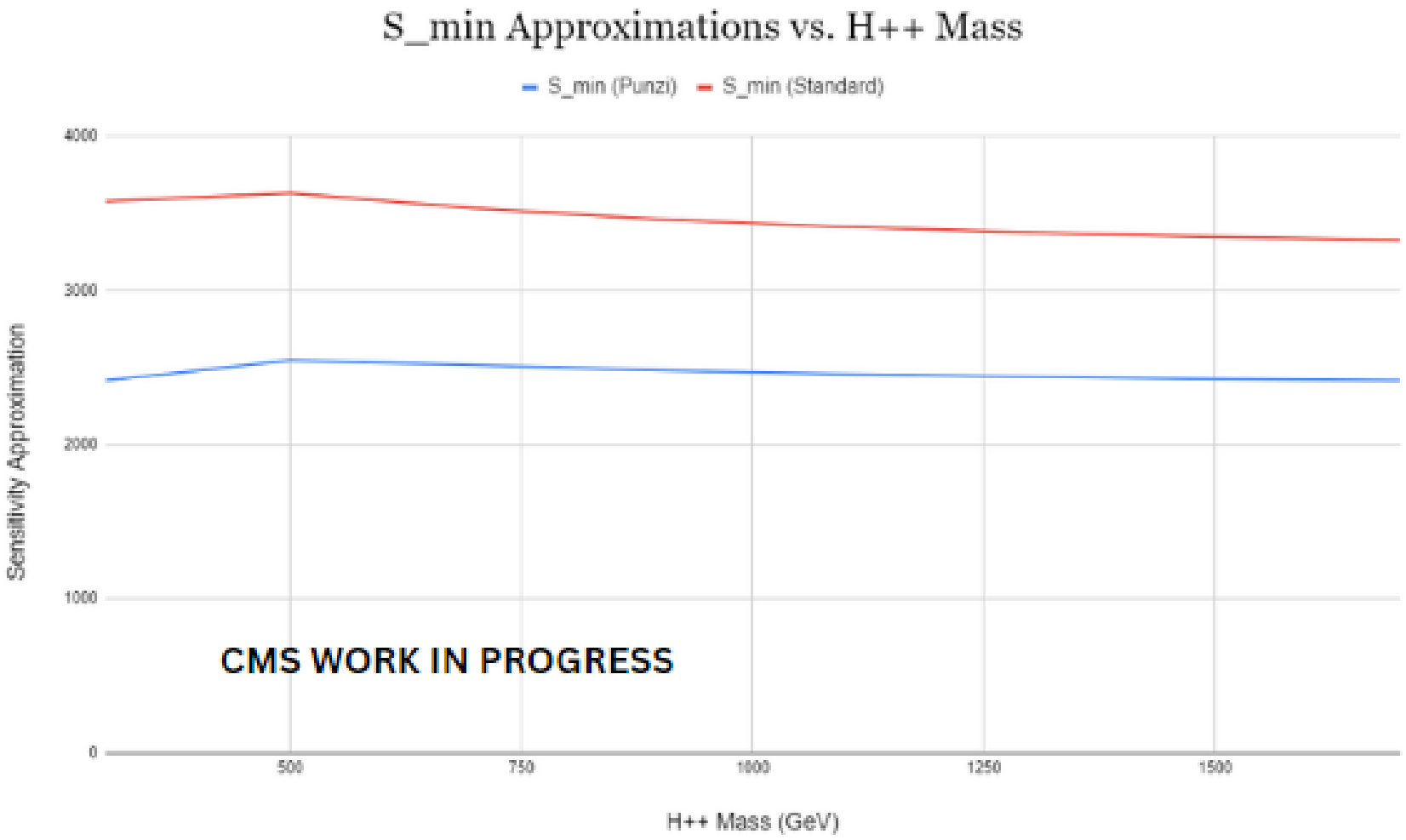
$$\frac{S}{S+B} \quad \text{and} \quad \frac{S}{\sqrt{B}}$$

- First expression cannot be calculated unless cross section of signal is known, and not directly related to significance of search, but rather the uncertainties in measurements of yields and new processes
- Second expression
  - Linear in S, but breaks down at small values of B
- If efficiencies are unknown

Some Limitations

- Cannot replace sensitivity search with full-likelihood approach
  - Only for optimization of analysis at early-stages of cut selection
  - Assumes a counting experiment and neglects systematics

$$\frac{S}{2.5 + \sqrt{B}}$$



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Future Work

- Background estimates
  - Photon fakes
  - Charge mismeasurement
  - Addressing other background sources
- Implement systematics into datacard generator
- Implement

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